Description

Image





Caption

1. Titanium Billet © Kirt Edblom at flickr (CC BY 2.0) 2. Titanium makes cool colors and patterns when it gets burned © David DeHetre at flickr (CC BY 2.0) 3. PowerBook G4 Titanium © Raneko at flickr (CC BY 2.0)

The material

Titanium is the seventh most abundant metal in the Earth's crust, but extracting the metal from the oxide in which it occurs naturally is unusually difficult. This makes titanium, third member of the light alloy trio, by far the most expensive of the three (more than ten times the price of aluminum). Despite this, the use of titanium is growing, propelled by its remarkable properties. It has a high melting point (1677 C), it is light, and - although reactive - its resists corrosion in most chemicals, protected by a thin film of oxide on its surface. Titanium has unusually low thermal and electrical conductivity, and low expansion coefficients.

Compositional summary

> 99% Ti

General properties

Density		4.51e3			kg/m^3
Price	*	11	-	14.7	GBP/kg
Date first used		1910			

Mechanical properties

Young's modulus	100	-	105	GPa
Shear modulus	36	-	39	GPa
Bulk modulus	111	-	135	GPa
Poisson's ratio	0.35	-	0.37	
Yield strength (elastic limit)	276	-	360	MPa
Tensile strength	345	-	490	MPa

Compressive strength	*	170	-	210	MPa
Elongation		20	-	26	% strain
Hardness - Vickers		155	-	165	HV
Fatigue strength at 10^7 cycles	*	245	-	296	MPa
Fracture toughness		55	-	60	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.002	-	0.003	

Thermal properties

Melting point		1.67e3			°C
Maximum service temperature	*	300	-	330	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Poor conductor			
Thermal conductivity		16.3	-	18	W/m.°C
Specific heat capacity		539	-	541	J/kg.°C
Thermal expansion coefficient		8.5	-	9.3	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		55	-	57.5	µohm.cm

Optical properties

Transparency	Opaque

Critical Materials Risk

High critical material risk?	Yes
Notes	Ti (titanium) added to the 2018 US critical minerals list

Processability

Castability	3			
Formability	2	-	4	
Machinability	1	-	3	
Weldability	4	-	5	
Solder/brazability	1	-	2	

Durability: water and aqueous solutions

Water (fresh)	Excellent
Water (salt)	Excellent
Soils, acidic (peat)	Excellent
Soils, alkaline (clay)	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Acceptable
Citric acid (10%)	Acceptable
Hydrochloric acid (10%)	Acceptable
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Limited use
Nitric acid (10%)	Excellent
Nitric acid (70%)	Excellent
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Acceptable
Sulfuric acid (10%)	Acceptable
Sulfuric acid (70%)	Limited use

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Limited use
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	

	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Limited use
O2 (oxygen gas)	Excellent
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Acceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	7.6e6	tonne/yr
Reserves, principal component	6.73e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 33.5	-	37	kg/kg
Embodied energy (virgin grade production)	* 558	-	616	MJ/kg
Water usage	* 104	-	115	l/kg

Material processing: energy

Metal casting, embodied energy	* 54.4	-	60.4	MJ/kg
Metal extrusion, hot, embodied energy	* 15.5	-	20.9	MJ/kg
Metal rolling and forging, embodied energy	* 8.26	-	11	MJ/kg

Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg
Metal powder forming, embodied energy	* 89	-	99	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 1.33e4	-	1.52e4	MJ/kg
Machining, coarse, embodied energy	* 16.8	-	19.1	MJ/kg
Machining, fine, embodied energy	* 21.2	-	25.1	MJ/kg
Machining, grinding, embodied energy	* 25.6	-	31.1	MJ/kg
Machining, laser, CO2, embodied energy	* 1.59	-	1.8	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.98	-	4.43	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 1.02	-	1.37	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.54	-	0.721	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 7.28	-	8.09	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 1.1e3	-	1.25e3	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.1	-	1.25	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.39	-	1.64	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.68	-	2.04	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.131	-	0.148	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 9.71	-	10.7	kg/kg
Embodied energy, recycling	* 379	-	419	MJ/kg
Recycle fraction in current supply	52			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Extracting titanium from its ores is very energy intensive. It can be recycled provided it is not contaminated with oxygen.

Supporting information

Design guidelines

Titanium is expensive, requiring vacuum processing to prevent take up of oxygen, which makes it brittle. But it is unusually strong, light and corrosion resistant, so much so that pure titanium can be implanted in the body to repair broken bones. The drive to miniaturize consumer electronics gives titanium a growing importance in product design. The casings of mobile phones and portable computers are now so thin that polymers cannot take the strain - they are not stiff or strong enough. The strength and low density of titanium makes it - despite its cost - an attractive replacement.

Technical notes

Titanium is an allotropic metal that can exist in two different crystal structures (iron is another). At room temperature it is hexagonal, called alpha-titanium. At 884 C it transforms to a body-centered cubic structure called beta-titanium. Alloying elements stabilize one or other of the two structures giving great scope for manipulating properties by alloying and heat treatment. See the record for Titanium alloys for more.

Typical uses

Chemical engineering, heat exchangers, bioengineering, medical applications including surgical implants, eyeglass frames, sports equipment such as golf clubs and bicycles, casings for mobile phones and portable computers.

Links

ProcessUniverse

Producers

Reference